

Data Cleaning and Quality Assessment

Homework 1 – Tabular data

1 Exercises (100 pts)

1. (50 pts) [Tidy Data from Excel](#) from *Cleaning Data for Effective Data Science*.
2. (50 pts) [Tidy Data from SQL](#) from *Cleaning Data for Effective Data Science*.

2 Extra credit (10 pts)

Some machine learning algorithms discard any instances with missing data. Write a script that artificially creates missing data in a CSV file. This can be used to evaluate how missing data can affect such a machine learning algorithm. Your script should have the following arguments:

- **Required:** path to the input CSV file.
- **Optional:** percent of instances, between 1 and 99, that should have missing data (in addition to already existing missing data). By default this should be 15.
- **Optional:** column number – in which column should values be replaced with **na**. By default, the missing values should be randomly distributed across all columns.
- **Optional:** path to the output CSV file. By default, **_na** should be appended to the name of the input file.

Provide a short documentation on how to run your script.

Submit

Upload to Canvas your solution, in a ZIP file.